

Lab Program 3

Title: Code a Dockerised Python or Node.js Application

STEP 1: Create a directory and navigate inside it

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab2$ cd ..  
BitterSteel@victus:~/Coursework/devops/lab_programs$ mkdir lab3  
BitterSteel@victus:~/Coursework/devops/lab_programs$ cd lab3
```

STEP 2: Create **package.json** file

\$ npm init -y

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ npm init -y  
Wrote to /home/BitterSteel/Coursework/devops/lab_programs/lab3/package.json:  
  
{  
  "name": "lab3",  
  "version": "1.0.0",  
  "description": "",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC"  
}
```

STEP 3: Install dependencies

\$ npm install express

STEP 4: Create **index.js** file

\$ nano index.js

Write the node application

```
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req,res) => {
    res.send("App running inside Docker container");
});

app.listen(port, () => {
    console.log(`server running at port ${port}`);
});
```

STEP 5: Write **Dockerfile**

\$ nano Dockerfile

```
GNU nano 7.2
FROM node:20-alpine

WORKDIR /app

COPY package*.json ./

RUN npm install express

COPY . .

EXPOSE 3000

CMD ["node", "index.js"]
```

Project Structure

lab3/

node_modules

index.js

Dockerfile

package-lock.json

package.json

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ ls
Dockerfile  index.js  node_modules  package.json  package-lock.json
```

STEP 6: Build Docker image

\$ docker build -t lab3:v1 .

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker build -t lab3:v1 .
[+] Building 2.3s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 170B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071
=> [internal] load build context
=> => transferring context: 43.71kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY package*.json ./
=> CACHED [4/5] RUN npm install express
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:ad458d72b50cd7f5c33ddf80eaedbda019b19e06731835d01f05fe723e82cd34
=> => naming to docker.io/library/lab3:v1
```

Verify the docker image

\$ docker images

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab2$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
lab3                 v1          ad458d72b50c  23 minutes ago 142MB
```

STEP 7: Run the docker container in detached mode, and bind the port

```
$ docker run -it -d -p 8080:3000 --name lab3-container lab3:v1
```

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker run -it -d -p 3000:3000 --name lab3.1-container lab3:v1
7b94502fc460bb826f5ec4fcbabbd5f7d05103bd2be5894b54f4ad4cabbcd5d5a
```

Check if container running

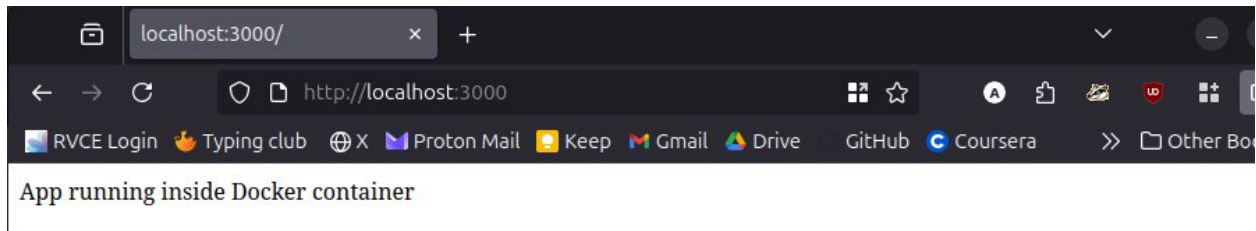
```
$ docker ps
```

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ff120e104f55	lab3:v1	"docker-entrypoint.s..."	15 seconds ago	Up 15 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	lab3.1-container

STEP 8: Access the application in browser

URL - <http://localhost:3000>



STEP 9: Stop the container

```
$ docker stop <container-id>
```

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker stop ff1
ff1
```

STEP 10: Remove the docker container & docker image

```
$ docker rm <container-id>
```

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker rm ff1
ff1
```

```
$ docker image rm <image-name>
```

```
BitterSteel@victus:~/Coursework/devops/lab_programs/lab3$ docker rmi lab3:v1
Untagged: lab3:v1
Deleted: sha256:ad458d72b50cd7f5c33ddf80eaedbda019b19e06731835d01f05fe723e82cd34
```